COVID-19 Analysis

2026-10-14

Introduction

The COVID-19 data is acquired from John Hopkins University Repository from the Center for Systems Science and Engineering. The end date for the data collection is 3/10/2023 and was aggregated from multiple different sources which can be referenced at the following location: https://github.com/CSSEGISandData/COVID-19. Four datasets are used, Global COVID-19 cases, global COVID-19 deaths, United States COVID-19 cases and United States COVID-19 deaths.

Research Questions

The purpose of this project is to analyze the patterns and relationships between reported COVID-19 cases and deaths in the United States and world wide using data collected by John Hopkings University. The analysis focuses on how cases and deaths chagned over time, how death rates differed across geographical locations, and the relationship between confirmed cases and reported deaths.

How did reported COVID-19 cases and deaths change over time in the United States and globally between 2020 and March 2023? How did reported COVID_19 death rates per 1000 people different amoung U.S. states and countries? What is the relationship between reported COVID-19 cases and deaths and how well can regression explain that relationship?

Data Load and Cleaning

Global Cases & Deaths

Loading four data sets and saving them for global cases & deaths and United States cases & deaths.

```
url_in <-"https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covi
file_names <- c("time_series_covid19_confirmed_global.csv",
                "time_series_covid19_deaths_global.csv",
                "time_series_covid19_confirmed_US.csv",
                "time_series_covid19_deaths_US.csv"
                )
urls <- str_c(url_in, file_names)
urls
```

```
## [1] "https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_
## [2] "https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_
## [3] "https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_
## [4] "https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_
```

Save CSV as variable

```
global_cases <- read.csv(urls[1])
global_deaths <- read.csv(urls[2])
US_cases <- read.csv(urls[3])
US_deaths <- read.csv(urls[4])
```

Filtering the global cases to exclude Latitude and Longitude. Un-pivoting the dates.

Global deaths filtering and un-pivot.

```
global_deaths <- global_deaths %>%
  select(-c(Lat, Long)) %>%
  pivot_longer(
    cols = -c(Province.State, Country.Region),
    names_to = "date",
    values_to = "deaths")
```

View column names

```
colnames(global_cases)
```

```
## [1] "Province.State" "Country.Region" "date"           "cases"
```

```
colnames(global_deaths)
```

```
## [1] "Province.State" "Country.Region" "date"           "deaths"
```

Joining the cases with the deaths by Province.State, Country.Region and date.

```
global <- global_cases %>%
  full_join(
    global_deaths,
    by = c("Province.State", "Country.Region", "date")
  ) %>%
  rename(
    country_region = Country.Region,
    province_state = Province.State
  ) %>%
  mutate(
    date = as.Date(
      sub("^X", "", date),
      format = "%m.%d.%y"
    )
  )

head(global)
```

```
## # A tibble: 6 x 5
##   province_state country_region date       cases deaths
##   <chr>          <chr>          <date>     <int>  <int>
## 1 ""             Afghanistan    2020-01-22     0      0
## 2 ""             Afghanistan    2020-01-23     0      0
```

```
## 3 ""             Afghanistan    2020-01-24     0      0
## 4 ""             Afghanistan    2020-01-25     0      0
## 5 ""             Afghanistan    2020-01-26     0      0
## 6 ""             Afghanistan    2020-01-27     0      0
```

View Summary of global dataset

```
summary(global)
```

```
##  province_state     country_region          date                cases
##  Length:330327      Length:330327      Min.   :2020-01-22   Min.   :        0
##  Class :character   Class :character   1st Qu.:2020-11-02   1st Qu.:      680
##  Mode  :character   Mode  :character   Median :2021-08-15   Median :    14429
##                                        Mean   :2021-08-15   Mean   :   959384
##                                        3rd Qu.:2022-05-28   3rd Qu.:   228517
##                                        Max.   :2023-03-09   Max.   :103802702
##      deaths
##  Min.   :      0
##  1st Qu.:      3
##  Median :    150
##  Mean   :  13380
##  3rd Qu.:   3032
##  Max.   :1123836
```

Limit global data set to records that have greater than 0 cases

```
global <- global %>% filter(cases > 0)
```

View records with large case numbers

```
global %>% filter(cases > 28000000)
```

```
## # A tibble: 2,510 x 5
##    province_state country_region date          cases deaths
##    <chr>          <chr>          <date>        <int>  <int>
##  1 ""             Brazil         2022-02-18 28072238 643340
##  2 ""             Brazil         2022-02-19 28177367 644195
##  3 ""             Brazil         2022-02-20 28218180 644592
##  4 ""             Brazil         2022-02-21 28258458 644918
##  5 ""             Brazil         2022-02-22 28361951 645735
##  6 ""             Brazil         2022-02-23 28493336 646714
##  7 ""             Brazil         2022-02-24 28589235 647703
##  8 ""             Brazil         2022-02-25 28679671 648496
##  9 ""             Brazil         2022-02-26 28749552 649184
## 10 ""             Brazil         2022-02-27 28776794 649437
## # i 2,500 more rows
```

United States Cases & Deaths

Filtering columns, and un-pivot dates on US cases.

```
us_cases = US_cases %>%
  select(-c(Lat, Long_)) %>%
  pivot_longer(
    cols =
      -c(UID:Combined_Key),
      names_to = "date",
      values_to = "cases"
      ) %>%
  select(Admin2:cases) %>%
  mutate(
    date = as.Date(
      sub("^X", "", date),
      format = "%m.%d.%y"
    )
    )
```

View first 5 records

```
head(us_cases)
```

```
## # A tibble: 6 x 6
##   Admin2  Province_State Country_Region Combined_Key         date       cases
##   <chr>   <chr>          <chr>          <chr>                <date>     <int>
## 1 Autauga Alabama        US             Autauga, Alabama, US 2020-01-22     0
## 2 Autauga Alabama        US             Autauga, Alabama, US 2020-01-23     0
## 3 Autauga Alabama        US             Autauga, Alabama, US 2020-01-24     0
## 4 Autauga Alabama        US             Autauga, Alabama, US 2020-01-25     0
## 5 Autauga Alabama        US             Autauga, Alabama, US 2020-01-26     0
## 6 Autauga Alabama        US             Autauga, Alabama, US 2020-01-27     0
```

Filtering and un-pivoting dates on US deaths.

```
us_deaths = US_deaths %>%
  select(-c(Lat, Long_)) %>%
  pivot_longer(
    cols =
      -c(UID:Population),
      names_to = "date",
      values_to = "cases"
      ) %>%
  select(Admin2:cases) %>%
  rename(deaths = cases) %>%
  mutate(
    date = as.Date(
      sub("^X", "", date),
      format = "%m.%d.%y"
    )
    )
```

View first 5 records in US deaths.

```r
head(us_deaths)
```

```
## # A tibble: 6 x 7
##   Admin2 Province_State Country_Region Combined_Key Population date       deaths
##   <chr>  <chr>          <chr>          <chr>             <int> <date>      <int>
## 1 Autau~ Alabama        US             Autauga, Al~      55869 2020-01-22      0
## 2 Autau~ Alabama        US             Autauga, Al~      55869 2020-01-23      0
## 3 Autau~ Alabama        US             Autauga, Al~      55869 2020-01-24      0
## 4 Autau~ Alabama        US             Autauga, Al~      55869 2020-01-25      0
## 5 Autau~ Alabama        US             Autauga, Al~      55869 2020-01-26      0
## 6 Autau~ Alabama        US             Autauga, Al~      55869 2020-01-27      0
```

Filter US deaths to exclude records with zero cases.

```r
us_cases %>%
  count(Province_State, Country_Region, date) %>%
  filter(n > 1)
```

```
## # A tibble: 59,436 x 4
##    Province_State Country_Region date           n
##    <chr>          <chr>          <date>     <int>
##  1 Alabama        US             2020-01-22    69
##  2 Alabama        US             2020-01-23    69
##  3 Alabama        US             2020-01-24    69
##  4 Alabama        US             2020-01-25    69
##  5 Alabama        US             2020-01-26    69
##  6 Alabama        US             2020-01-27    69
##  7 Alabama        US             2020-01-28    69
##  8 Alabama        US             2020-01-29    69
##  9 Alabama        US             2020-01-30    69
## 10 Alabama        US             2020-01-31    69
## # i 59,426 more rows
```

Filter US deaths to exclude records with zero cases.

```r
us_deaths %>%
  count(Province_State, Country_Region, date) %>%
  filter(n > 1)
```

```
## # A tibble: 59,436 x 4
##    Province_State Country_Region date           n
##    <chr>          <chr>          <date>     <int>
##  1 Alabama        US             2020-01-22    69
##  2 Alabama        US             2020-01-23    69
##  3 Alabama        US             2020-01-24    69
##  4 Alabama        US             2020-01-25    69
##  5 Alabama        US             2020-01-26    69
##  6 Alabama        US             2020-01-27    69
##  7 Alabama        US             2020-01-28    69
##  8 Alabama        US             2020-01-29    69
##  9 Alabama        US             2020-01-30    69
## 10 Alabama        US             2020-01-31    69
## # i 59,426 more rows
```

Create US data set by joining US cases & deaths.

```
us <- us_cases %>%
  full_join(
    us_deaths,
    by = c("Province_State", "Country_Region", "date", "Admin2", "Combined_Key")
  ) %>%
  rename(
    country_region = Country_Region,
    province_state = Province_State
  ) %>%
  mutate(
    date = as.Date(date, format = "%m.%d.%y")

  )

head(us)
```

```
## # A tibble: 6 x 8
##   Admin2  province_state country_region Combined_Key date       cases Population
##   <chr>   <chr>          <chr>          <chr>        <date>     <int>      <int>
## 1 Autauga Alabama        US             Autauga, Al~ 2020-01-22     0      55869
## 2 Autauga Alabama        US             Autauga, Al~ 2020-01-23     0      55869
## 3 Autauga Alabama        US             Autauga, Al~ 2020-01-24     0      55869
## 4 Autauga Alabama        US             Autauga, Al~ 2020-01-25     0      55869
## 5 Autauga Alabama        US             Autauga, Al~ 2020-01-26     0      55869
## 6 Autauga Alabama        US             Autauga, Al~ 2020-01-27     0      55869
## # i 1 more variable: deaths <int>
```

Create the "combined_Key" feature using province_state and country_region.

```
global <- global %>%
  unite("Combined_Key",
        c("province_state", "country_region"),
        sep = ", ",
        na.rm = TRUE,
        remove = FALSE
        )
```

The US data features population. For consistency, adding population to the global dataset using the"Combined_Key" to join.

```
uid_lookup_url <- "https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/U
```

```
uid = read_csv(uid_lookup_url) %>%
  select(-c(Lat, Long_, Combined_Key, code3, iso2, iso3, Admin2))
```

```
## Rows: 4321 Columns: 12
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (7): iso2, iso3, FIPS, Admin2, Province_State, Country_Region, Combined_Key
## dbl (5): UID, code3, Lat, Long_, Population
```

```
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Renaming columns to match joining data sets.

```
uid = uid %>%
  rename(
    country_region = Country_Region,
    province_state = Province_State
  )
```

Summarizing statistics in the United States by state.

```
us_by_state <- us %>%
  group_by(province_state, country_region, date) %>%
  summarize(cases = sum(cases), deaths = sum(deaths),
            Population = sum(Population)) %>%
  mutate(deaths_per_mill = deaths * 1000000 / Population) %>%
  select(province_state, country_region, date, cases, deaths, deaths_per_mill, Population) %>%
  ungroup()
```

```
## `summarise()` has grouped output by 'province_state', 'country_region'. You can
## override using the `.groups` argument.
```

```
tail(us_by_state)
```

```
## # A tibble: 6 x 7
##   province_state country_region date        cases deaths deaths_per_mill
##   <chr>          <chr>          <date>      <int>  <int>           <dbl>
## 1 Wyoming        US             2023-03-04 185159   2002           3459.
## 2 Wyoming        US             2023-03-05 185159   2002           3459.
## 3 Wyoming        US             2023-03-06 185159   2002           3459.
## 4 Wyoming        US             2023-03-07 185385   2004           3463.
## 5 Wyoming        US             2023-03-08 185385   2004           3463.
## 6 Wyoming        US             2023-03-09 185385   2004           3463.
## # i 1 more variable: Population <int>
```

Summarizing statistics for the United States

```
us_total <- us_by_state %>%
  group_by(country_region, date) %>%
  summarize(cases = sum(cases), deaths = sum(deaths),
            Population = sum(Population)) %>%
  mutate(deaths_per_mill = deaths * 1000000 / Population) %>%
  select(country_region, date, cases, deaths, deaths_per_mill, Population) %>%
  ungroup()
```

```
## `summarise()` has grouped output by 'country_region'. You can override using
## the `.groups` argument.
```

```
tail(us_total)
```

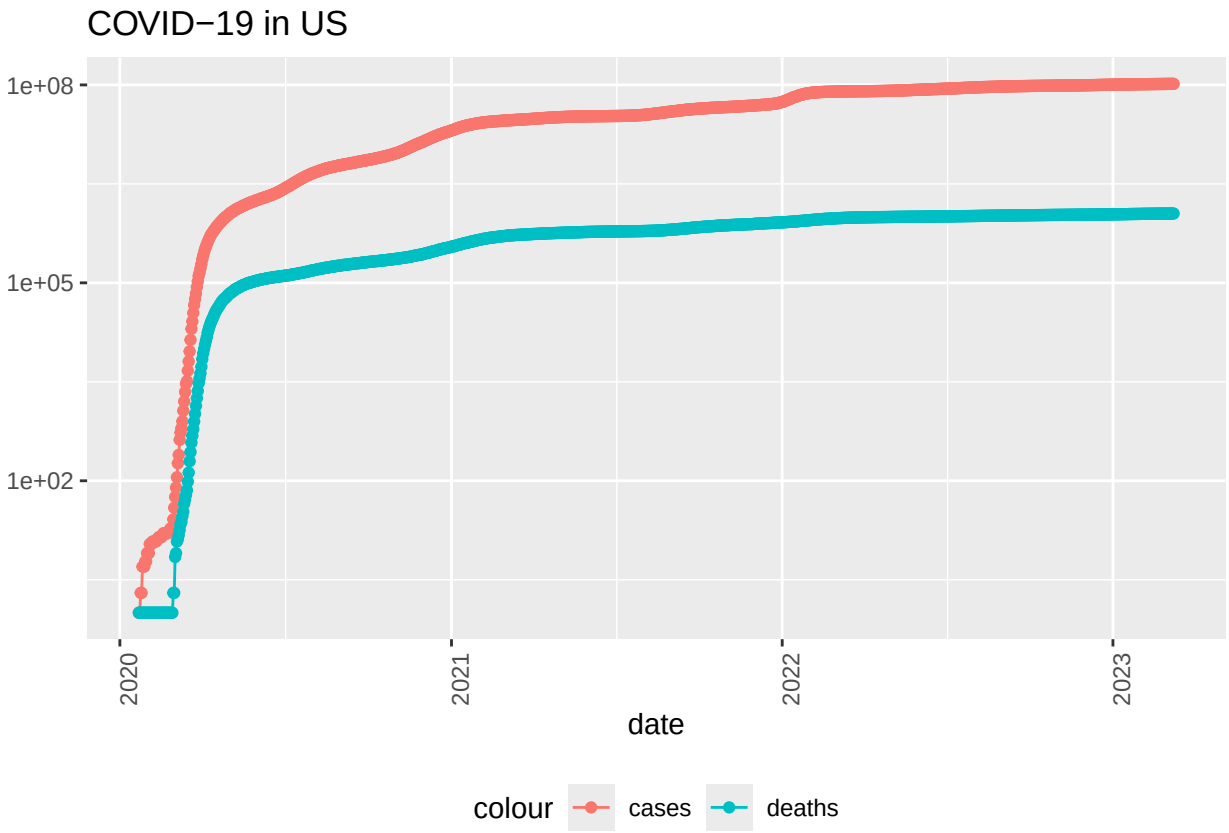
```
## # A tibble: 6 x 6
##   country_region date           cases  deaths deaths_per_mill Population
##   <chr>          <date>         <int>   <int>           <dbl>      <int>
## 1 US             2023-03-04 103650837 1122172           3371.  332875137
## 2 US             2023-03-05 103646975 1122134           3371.  332875137
## 3 US             2023-03-06 103655539 1122181           3371.  332875137
## 4 US             2023-03-07 103690910 1122516           3372.  332875137
## 5 US             2023-03-08 103755771 1123246           3374.  332875137
## 6 US             2023-03-09 103802702 1123836           3376.  332875137
```

Visualizations & Analysis

United States: COVID-19 Cases and Deaths

Plotting the cases and Deaths in the United States

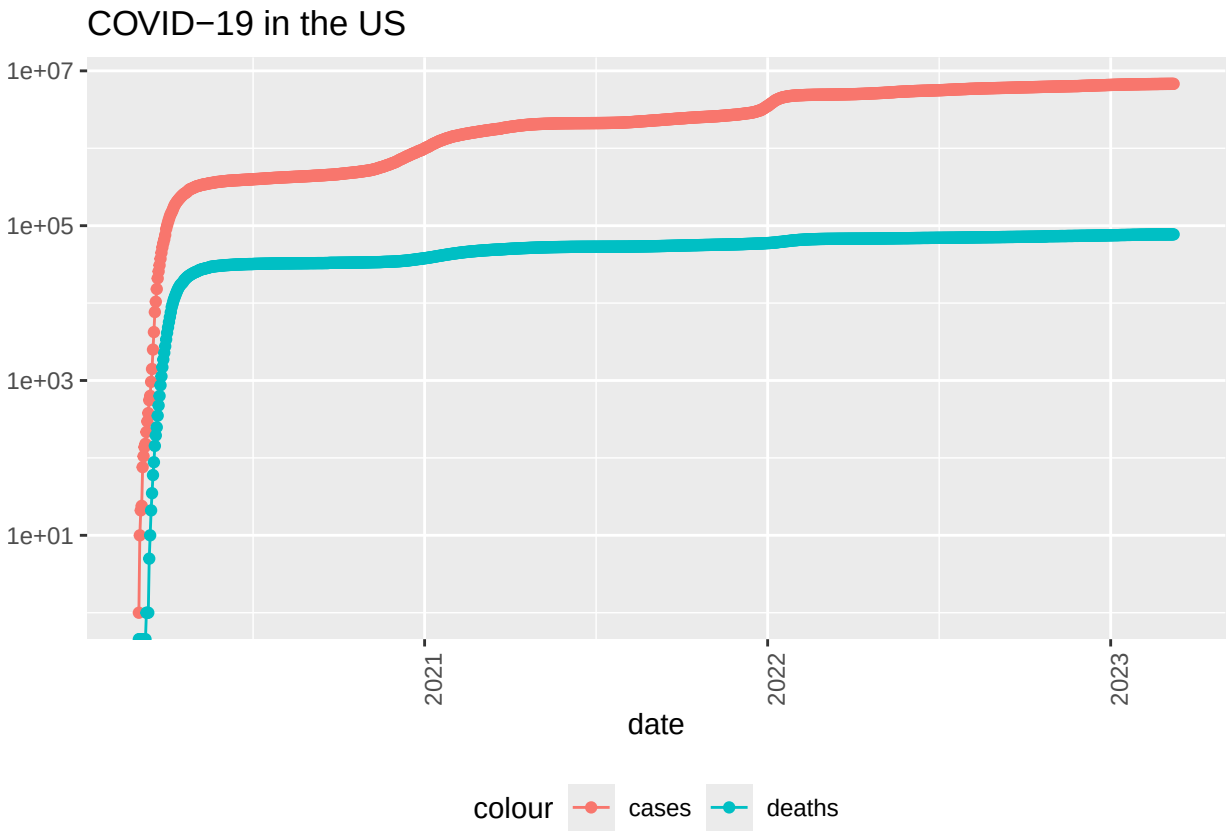
```
us_total %>%
  filter(cases > 0) %>%
  ggplot(aes(x = date, y = cases)) +
  geom_line(aes(color= "cases")) +
  geom_point(aes(color = "cases")) +
  geom_line(aes(y = deaths, color = "deaths")) +
  geom_point(aes(y = deaths, color = "deaths")) +
  scale_y_log10() +
  theme(legend.position = "bottom", axis.text.x = element_text(angle = 90)) +
  labs(title = "COVID-19 in US", y =NULL)
```



Plotting COVID-19 cases and deaths in New York

```
state = "New York"
us_by_state %>%
  filter(province_state == state) %>%
  filter(cases > 0) %>%
  ggplot(aes(x = date, y = cases)) +
  geom_line(aes(color= "cases")) +
  geom_point(aes(color = "cases")) +
  geom_line(aes(y = deaths, color = "deaths")) +
  geom_point(aes(y = deaths, color = "deaths")) +
  scale_y_log10() +
  theme(legend.position = "bottom", axis.text.x = element_text(angle = 90)) +
  labs(title = "COVID-19 in the US", y =NULL)
```

```
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
```




```
max(us_total$date)
```

```
## [1] "2023-03-09"
```

```
max(us_total$deaths)
```

```
## [1] 1123836
```

Calculating new cases and new deaths instead of cumulative rates.

```
us_by_state <- us_by_state %>%
  group_by(province_state) %>%
  arrange(date, .by_group = TRUE) %>%
  mutate(
    new_cases = cases - lag(cases),
    new_deaths = deaths - lag(deaths)
  ) %>%
  ungroup()

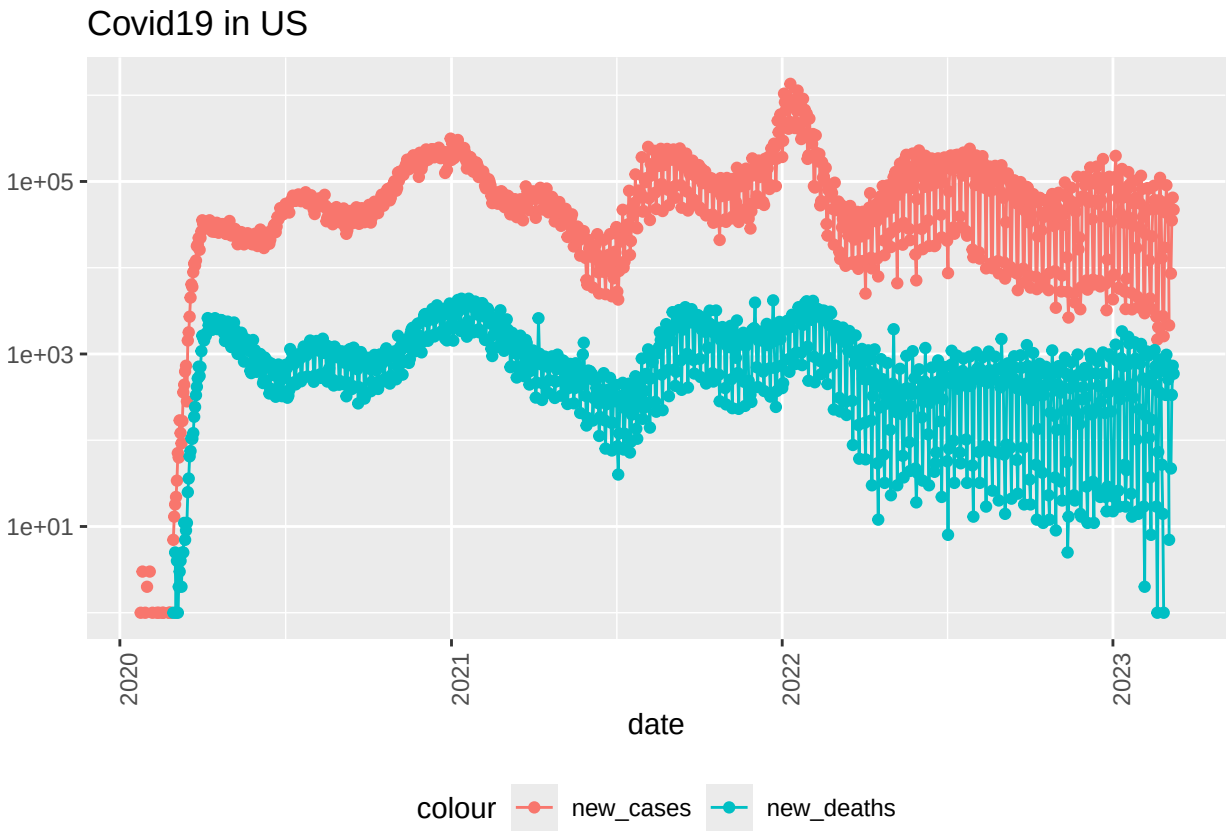
us_total = us_total %>%
  mutate(new_cases = cases - lag(cases),
         new_deaths = deaths - lag(deaths))
```

```
tail(us_total)
```

```
## # A tibble: 6 x 8
##   country_region date          cases deaths deaths_per_mill Population new_cases
##   <chr>          <date>        <int>  <int>           <dbl>      <int>     <int>
## 1 US             2023-03-04   1.04e8 1.12e6           3371.  332875137      2147
## 2 US             2023-03-05   1.04e8 1.12e6           3371.  332875137     -3862
## 3 US             2023-03-06   1.04e8 1.12e6           3371.  332875137      8564
## 4 US             2023-03-07   1.04e8 1.12e6           3372.  332875137     35371
## 5 US             2023-03-08   1.04e8 1.12e6           3374.  332875137     64861
## 6 US             2023-03-09   1.04e8 1.12e6           3376.  332875137     46931
## # i 1 more variable: new_deaths <int>
```

Plotting New Cases and Deaths in the United States

```
us_total %>%
  mutate(
    new_cases = ifelse(new_cases > 0, new_cases, NA),
    new_deaths = ifelse(new_deaths > 0, new_deaths, NA)
  ) %>%
  ggplot(aes(x = date, y = new_cases)) +
  geom_line(aes(color = "new_cases"), na.rm = TRUE) +
  geom_point(aes(color = "new_cases"), na.rm = TRUE) +
  geom_line(aes(y = new_deaths, color = "new_deaths"), na.rm = TRUE) +
  geom_point(aes(y = new_deaths, color = "new_deaths"), na.rm = TRUE) +
  scale_y_log10() +
  theme(
    legend.position = "bottom",
    axis.text.x = element_text(angle = 90)
  ) +
  labs(title = "Covid19 in US", y = NULL)
```



Plotting new cases and deaths in New York

```
state = "New York"
us_by_state %>%
  filter(province_state == state) %>%
  ggplot(aes(x = date, y = new_cases)) +
  geom_line(aes(color= "new_cases"), na.rm = TRUE) +
  geom_point(aes(color = "new_cases"), na.rm = TRUE) +
  geom_line(aes(y = new_deaths, color = "new_deaths"), na.rm = TRUE) +
  geom_point(aes(y = new_deaths, color = "new_deaths"), na.rm = TRUE) +
  scale_y_log10() +
  theme(legend.position = "bottom", axis.text.x = element_text(angle = 90)) +
  labs(title = "COVID-19 in New York", y =NULL)
```

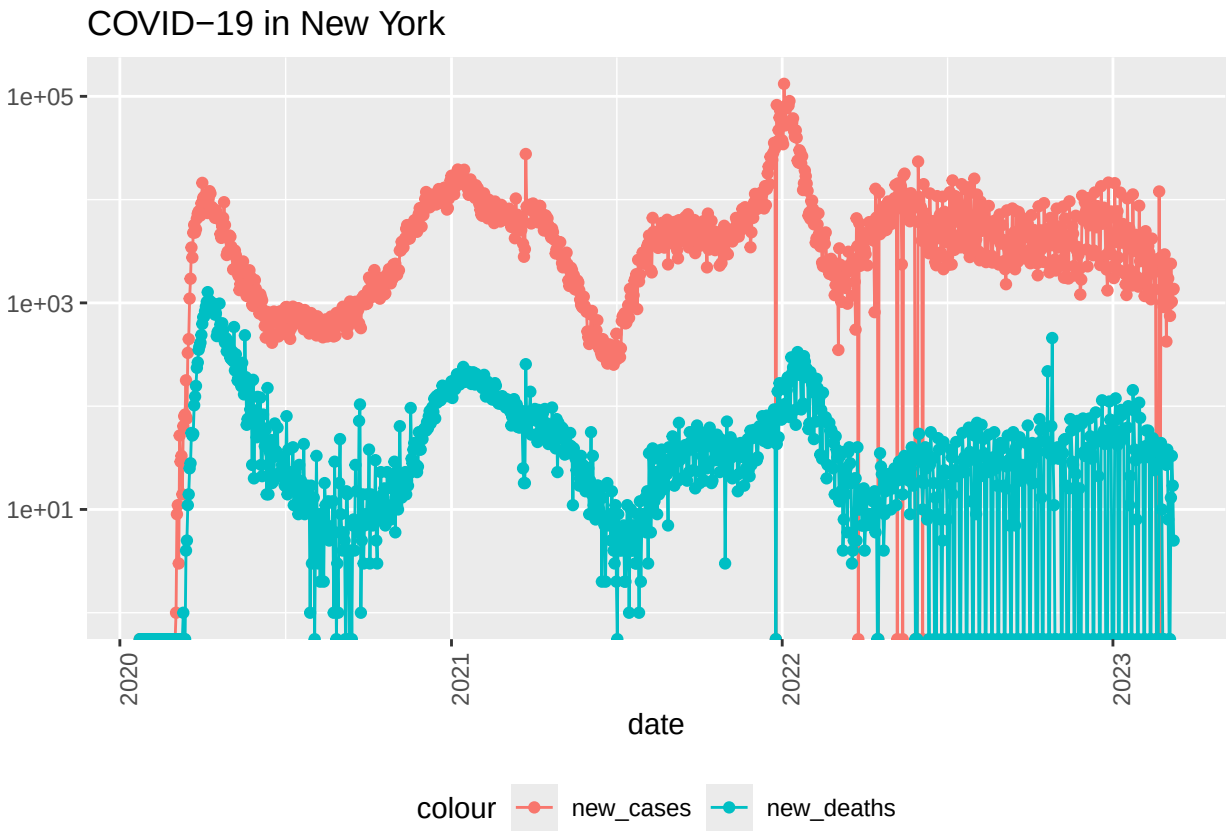
```
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.

## Warning in transformation$transform(x): NaNs produced

## Warning in scale_y_log10(): log-10 transformation introduced infinite values.

## Warning in transformation$transform(x): NaNs produced

## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
```



Summarizing statistics for State totals.

```
us_state_totals = us_by_state %>%
  group_by(province_state) %>%
  summarize(
    deaths = max(deaths),
    cases = max(cases),
    population = max(Population),
    cases_per_thou = (1000*cases / population),
    deaths_per_thou = (1000*deaths / population)) %>%
  filter(cases > 0, population > 0)
```

```
us_state_totals %>%
  slice_min(deaths_per_thou, n =10) %>%
  select(deaths_per_thou, cases_per_thou, everything())
```

```
## # A tibble: 10 x 6
##    deaths_per_thou cases_per_thou province_state        deaths  cases population
##              <dbl>          <dbl> <chr>                 <int>  <int>      <int>
##  1           0.611           150. American Samoa           34 8.32e3      55641
##  2           0.744           248. Northern Mariana Isl~    41 1.37e4      55144
##  3           1.21            231. Virgin Islands          130 2.48e4     107268
##  4           1.30            269. Hawaii                 1841 3.81e5    1415872
##  5           1.49            245. Vermont                 929 1.53e5     623989
##  6           1.55            293. Puerto Rico            5823 1.10e6    3754939
##  7           1.65            340. Utah                   5298 1.09e6    3205958
##  8           2.01            415. Alaska                 1486 3.08e5     740995
```

```
##  9          2.03           252. District of Columbia    1432 1.78e5     705749
## 10          2.06           253. Washington             15683 1.93e6    7614893
```

```
us_state_totals %>%
  slice_max(deaths_per_thou, n =10) %>%
  select(deaths_per_thou, cases_per_thou, everything())
```

```
## # A tibble: 10 x 6
##    deaths_per_thou cases_per_thou province_state deaths   cases population
##              <dbl>          <dbl> <chr>           <int>   <int>      <int>
##  1            4.55           336. Arizona         33102 2443514    7278717
##  2            4.54           326. Oklahoma        17972 1290929    3956971
##  3            4.49           333. Mississippi     13370  990756    2976149
##  4            4.44           359. West Virginia    7960  642760    1792147
##  5            4.32           320. New Mexico       9061  670929    2096829
##  6            4.31           334. Arkansas        13020 1006883    3017804
##  7            4.29           335. Alabama         21032 1644533    4903185
##  8            4.28           368. Tennessee       29263 2515130    6829174
##  9            4.23           307. Michigan        42205 3064125    9986857
## 10            4.06           385. Kentucky        18130 1718471    4467673
```

Analysis: COVID-19 Trends in the United States Over the course of the pandemic, cases and deaths spiked and then fluctuated. Peaks occurred around the new years of 2021 and 2022, showing seasonal trends. Arizona showed the highest deaths per thousand, while United States island territories and states such as American Somoa, North Mariana Island, Virgin Islands and Hawaii showed the lowest deaths per thousands. After Data may have been reported differently through out the data collection. The new deaths trend with the new cases, and both become more variable especially after 2022. After 2022, both variables have trended negatively signifying lowering rates of cases and deaths.

Global: COVID-19 Cases and Deaths

```
global
```

```
## # A tibble: 306,827 x 6
##    Combined_Key  province_state country_region date       cases deaths
##    <chr>         <chr>          <chr>          <date>     <int>  <int>
##  1 , Afghanistan ""             Afghanistan    2020-02-24     5      0
##  2 , Afghanistan ""             Afghanistan    2020-02-25     5      0
##  3 , Afghanistan ""             Afghanistan    2020-02-26     5      0
##  4 , Afghanistan ""             Afghanistan    2020-02-27     5      0
##  5 , Afghanistan ""             Afghanistan    2020-02-28     5      0
##  6 , Afghanistan ""             Afghanistan    2020-02-29     5      0
##  7 , Afghanistan ""             Afghanistan    2020-03-01     5      0
##  8 , Afghanistan ""             Afghanistan    2020-03-02     5      0
##  9 , Afghanistan ""             Afghanistan    2020-03-03     5      0
## 10 , Afghanistan ""             Afghanistan    2020-03-04     5      0
## # i 306,817 more rows
```

```
global_country_totals = global %>%
  group_by(country_region) %>%
  summarize(
    deaths = max(deaths),
    cases = max(cases),
    ) %>%
  filter(cases > 0)
```

```
global_country_totals <- global_country_totals %>%
  left_join(uid, by = c("country_region")) %>%
  select(-c(UID, FIPS)) %>%
  select(country_region, cases, deaths, Population)
```

```
uid_country <- uid %>%
  mutate(
    Population = as.numeric(Population)
  ) %>%
  filter(
    is.na(province_state) | province_state == ""
  ) %>%
  group_by(country_region) %>%
  summarize(
    population = if (all(is.na(Population))) {
      NA_real_
    } else {
      max(Population, na.rm = TRUE)
    },
    .groups = "drop"
  )
```

```
global_country_totals <- global %>%
  group_by(country_region, date) %>%
  summarize(
    cases = sum(cases, na.rm = TRUE),
    deaths = sum(deaths, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  group_by(country_region) %>%
  filter(date == max(date)) %>%
  ungroup() %>%
  left_join(uid_country, by = "country_region") %>%
  filter(
    !is.na(population),
    population > 0
  ) %>%
  mutate(
    cases_per_mil = cases / population * 1000000,
    deaths_per_mil = deaths / population * 1000000
  )
```

```
global_country_totals %>%
  select(
    country_region,
```

```
    population,
    cases,
    deaths,
    deaths_per_mil
  ) %>%
  head(10)
```

```
## # A tibble: 10 x 5
##    country_region      population    cases deaths deaths_per_mil
##    <chr>                    <dbl>    <int>  <int>          <dbl>
##  1 Afghanistan           38928341   209451   7896          203.
##  2 Albania                2877800   334457   3598         1250.
##  3 Algeria               43851043   271496   6881          157.
##  4 Andorra                  77265    47890    165         2136.
##  5 Angola                32866268   105288   1933           58.8
##  6 Antigua and Barbuda      97928     9106    146         1491.
##  7 Argentina             45195777 10044957 130472         2887.
##  8 Armenia                2963234   447308   8727         2945.
##  9 Australia             25459700 11399460  19574          769.
## 10 Austria                9006400  5961143  21970         2439.
```

```
global_totals <- global %>%
  group_by(country_region, date) %>%
  summarize(
    cases = sum(cases, na.rm = TRUE),
    deaths = sum(deaths, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  group_by(country_region) %>%
  filter(date == max(date)) %>%
  ungroup()
```

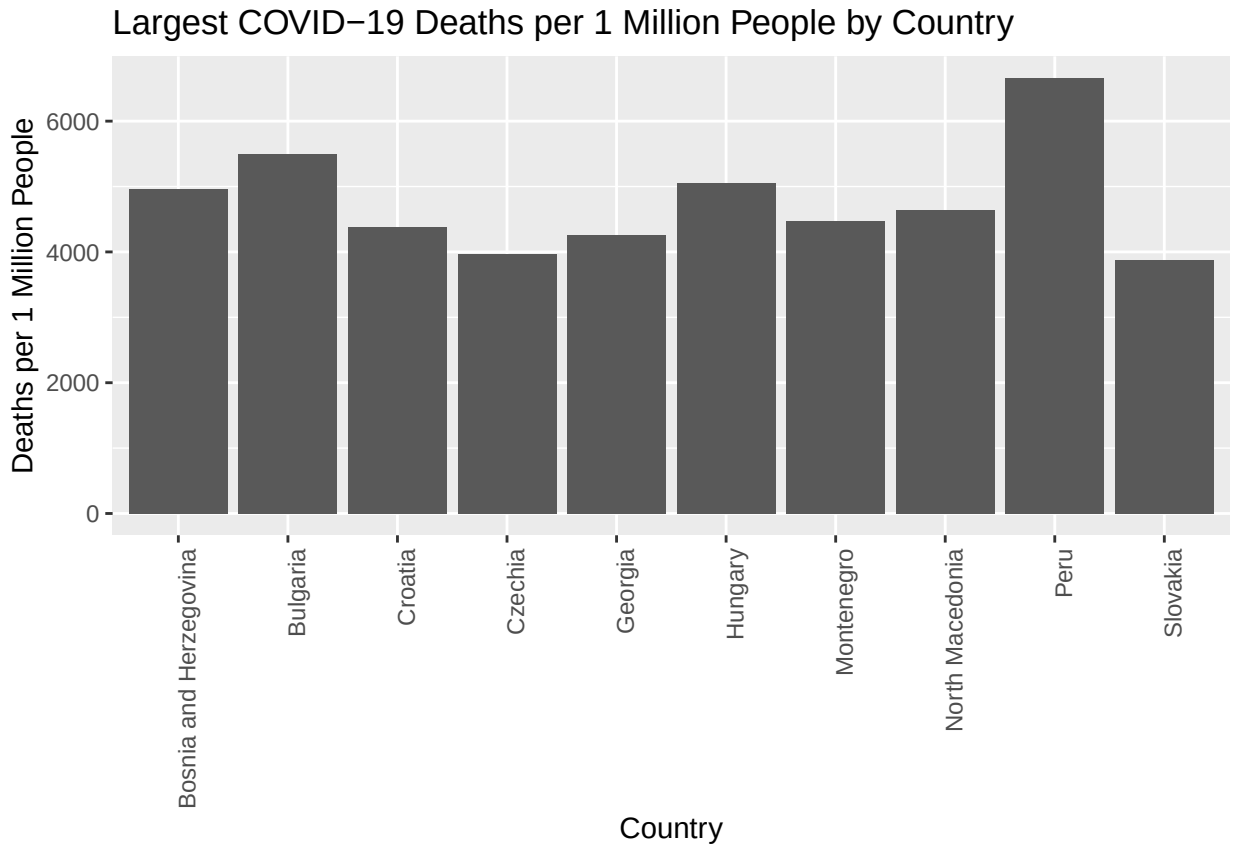
```
largest_global = global_country_totals %>%
  slice_max(deaths_per_mil, n =10) %>%
  select(deaths_per_mil, everything())
```

```
largest_global
```

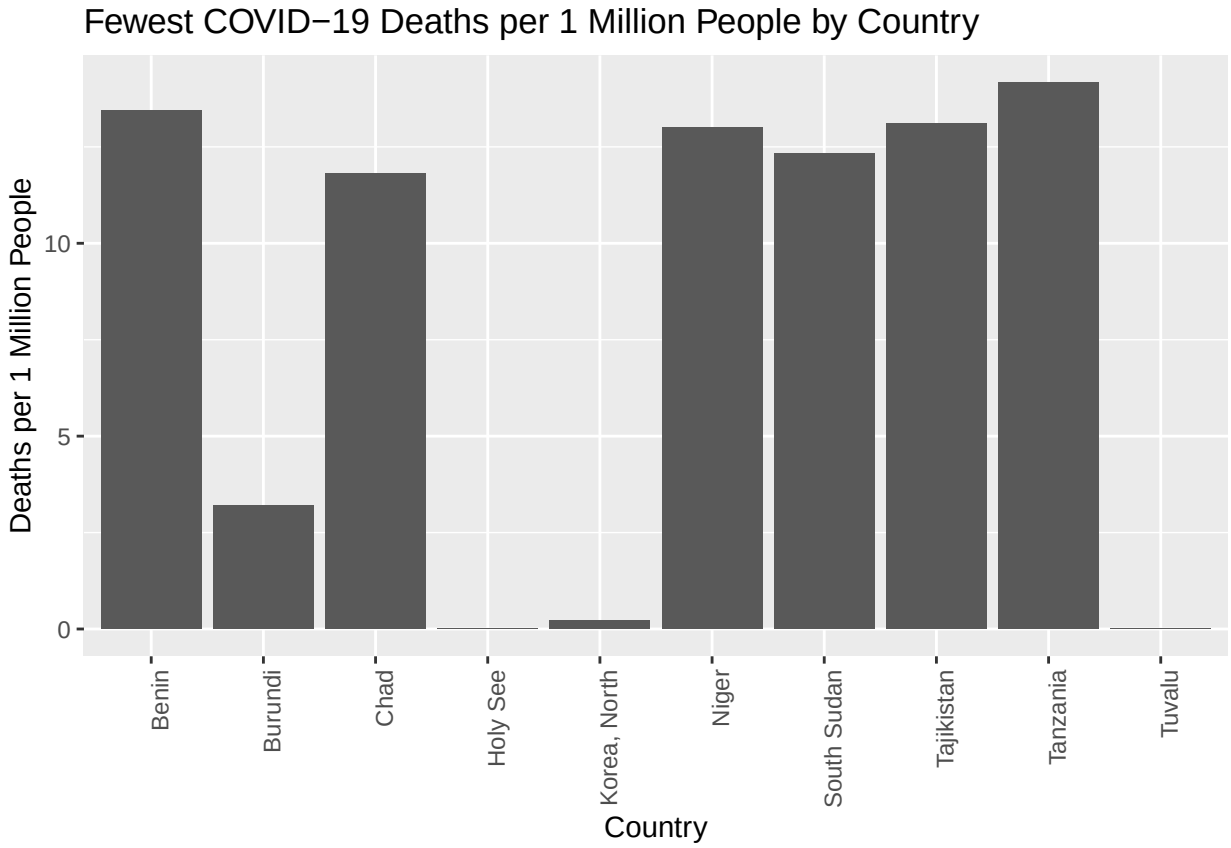
```
## # A tibble: 10 x 7
##    deaths_per_mil country_region         date         cases deaths population
##             <dbl> <chr>                  <date>       <int>  <int>      <dbl>
##  1          6658. Peru                   2023-03-09 4487553 219539   32971846
##  2          5502. Bulgaria               2023-03-09 1297523  38228    6948445
##  3          5048. Hungary                2023-03-09 2196804  48762    9660350
##  4          4962. Bosnia and Herzegovina 2023-03-09  401729  16280    3280815
##  5          4638. North Macedonia        2023-03-09  346852   9662    2083380
##  6          4471. Montenegro             2023-03-09  288808   2808     628062
##  7          4381. Croatia                2023-03-09 1269326  17987    4105268
##  8          4254. Georgia                2023-03-09 1827537  16971    3989175
##  9          3968. Czechia                2023-03-09 4618256  42491   10708982
## 10          3870. Slovakia               2023-03-09 2667551  21035    5434712
## # i 1 more variable: cases_per_mil <dbl>
```

```
smallest_global = global_country_totals %>%
  slice_min(deaths_per_mil, n =10) %>%
  select(deaths_per_mil, everything())
```

```
ggplot(largest_global, aes(x = country_region, y = deaths_per_mil)) +
  geom_col() +
  labs(
    x = "Country",
    y = "Deaths per 1 Million People",
    title = "Largest COVID-19 Deaths per 1 Million People by Country"
  ) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1))
```




```
ggplot(smallest_global, aes(x = country_region, y = deaths_per_mil)) +
  geom_col() +
  labs(
    x = "Country",
    y = "Deaths per 1 Million People",
    title = "Fewest COVID-19 Deaths per 1 Million People by Country"
  ) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1))
```



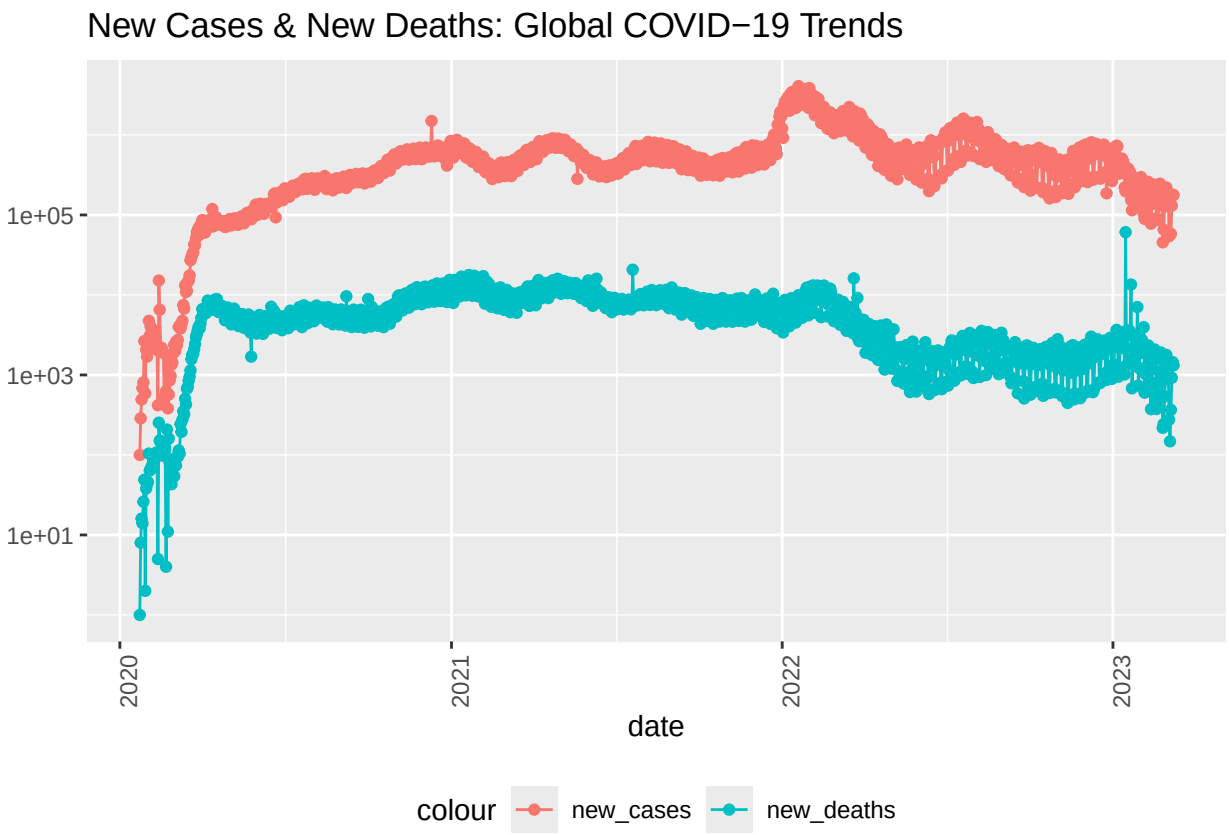

```
global_total <- global %>%
  group_by(date) %>%
  summarize(
    cases = sum(cases, na.rm = TRUE),
    deaths = sum(deaths, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(date) %>%
  mutate(
    new_cases = cases - lag(cases),
    new_deaths = deaths - lag(deaths)
  )
```

```
global_total
```

```
## # A tibble: 1,143 x 5
##    date       cases deaths new_cases new_deaths
##    <date>     <int>  <int>     <int>      <int>
##  1 2020-01-22   557     17        NA         NA
##  2 2020-01-23   657     18       100          1
##  3 2020-01-24   944     26       287          8
##  4 2020-01-25  1437     42       493         16
##  5 2020-01-26  2120     56       683         14
##  6 2020-01-27  2929     82       809         26
##  7 2020-01-28  5580    131      2651         49
##  8 2020-01-29  6169    133       589          2
```

```
##  9 2020-01-30  8237    171      2068         38
## 10 2020-01-31  9927    214      1690         43
## # i 1,133 more rows
```

```
global_total %>%
  mutate(
    new_cases = ifelse(new_cases > 0, new_cases, NA),
    new_deaths = ifelse(new_deaths > 0, new_deaths, NA)
  ) %>%
  ggplot(aes(x = date, y = new_cases)) +
  geom_line(aes(color = "new_cases"), na.rm = TRUE) +
  geom_point(aes(color = "new_cases"), na.rm = TRUE) +
  geom_line(aes(y = new_deaths, color = "new_deaths"), na.rm = TRUE) +
  geom_point(aes(y = new_deaths, color = "new_deaths"), na.rm = TRUE) +
  scale_y_log10() +
  theme(
    legend.position = "bottom",
    axis.text.x = element_text(angle = 90)
  ) +
  labs(title = "New Cases & New Deaths: Global COVID-19 Trends", y = NULL)
```




```
global_yearly <- global_total %>%
  arrange(date) %>%
  mutate(
    year = lubridate::year(date),
    new_cases = cases - lag(cases),
    new_deaths = deaths - lag(deaths)
```

```
  ) %>%
  filter(!is.na(new_cases), !is.na(new_deaths)) %>%
  group_by(year) %>%
  summarise(
    annual_cases = sum(new_cases),
    annual_deaths = sum(new_deaths),
    .groups = "drop"
  ) %>%
  mutate(
    death_percentage = 100 * annual_deaths / annual_cases
  )

global_yearly
```

```
## # A tibble: 4 x 4
##    year annual_cases annual_deaths death_percentage
##   <dbl>        <int>         <int>            <dbl>
## 1  2020     83778065       1901759            2.27
## 2  2021    204949669       3567242            1.74
## 3  2022    371735501       1224357            0.329
## 4  2023     16106357        188426            1.17
```

Analysis: COVID-19 Trends across the Globe The global trends of COVID-19 cases and deaths across the globe are similar to the United States. There was a sharp spike in 2020, followed by a leveling out. While new cases still climbed in a relatively shallow way, the new death cases remained relatively flat. Larger variance is see after 2022-2023. At the end of data collection in 2023, new cases and deaths were at or near the highest levels of 2020, recovering from any increases in 2021-2022. The countries showing the largest deaths per million where Peru and many countries in Eastern Europe. The countries show the fewest deaths per million were Holy See and North Korea. Holy See, being the Vatican enclave in Rome, has 1,000 person population and zero COVID-19 deaths as of the data collection.

Annual tends show the death percentage decreasing by over a full percent between 2020 and 2023.

Modeling

United States Multiple Linear Regression Model

```
mod = lm(deaths_per_thou ~ + cases_per_thou + log(population), data = us_state_totals)
```

```
summary(mod)
```

```
##
## Call:
## lm(formula = deaths_per_thou ~ +cases_per_thou + log(population),
##     data = us_state_totals)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.79962 -0.40913  0.06935  0.57162  1.18912
##
```

```
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)     -4.380037   1.147188  -3.818 0.000354 ***
## cases_per_thou   0.009346   0.002083   4.488 3.91e-05 ***
## log(population)  0.310635   0.073927   4.202 0.000102 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7532 on 53 degrees of freedom
## Multiple R-squared:  0.4795, Adjusted R-squared:  0.4599
## F-statistic: 24.41 on 2 and 53 DF,  p-value: 3.056e-08
```

```
x_grid = seq(1, 151)
```

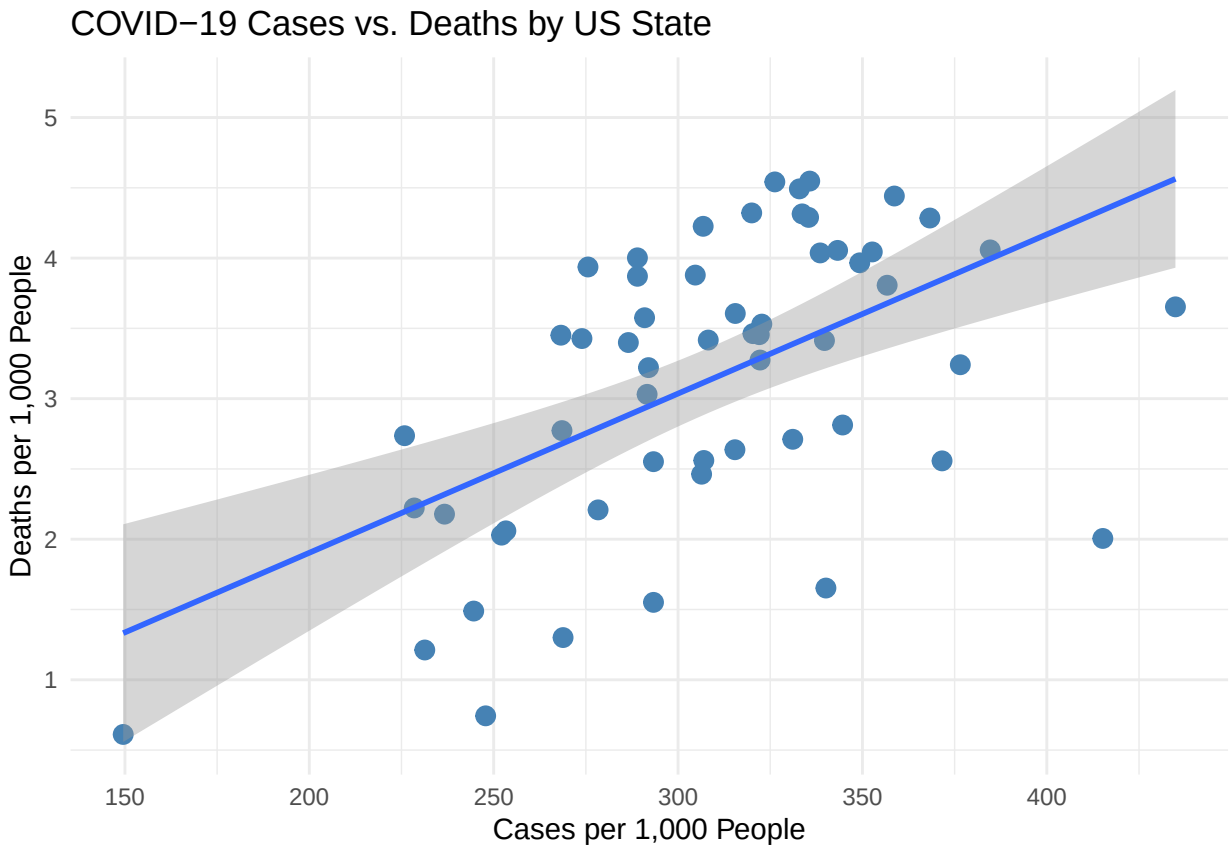
```
new_df = tibble(cases_pre_thou = x_grid)
```

```
us_tot_pred = us_state_totals %>% mutate(pred = predict(mod))
```

Plotting predicted deaths from cases vs true deaths and cases.

```
us_tot_pred %>%
  ggplot(aes(x = cases_per_thou, y = deaths_per_thou)) +
  geom_point(color = "steelblue", size = 3) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "COVID-19 Cases vs. Deaths by US State",
    x = "Cases per 1,000 People",
    y = "Deaths per 1,000 People"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



####Analysis: United States Multiple Linear Regression Model

The multiple linear regression model predicts deaths per thousand using cases per thousand and the log of the population. The deaths per thousand were calculated on a lag behind the cases. The Adjusted R-squared is 0.4599, showing that the model explains approximately 46% of the variation. The p-values show that both the cases per thousand and population are significant. However, the pattern in the scatter plot may show some non-linearity that the model does not capture. Also, as time passes, treatments improve and date of infection may also play a role in the deaths per thousand prediction.

Global COVID-19 Linear Regression & Polynomial Regression

```
global_mod <- global_country_totals %>%
  filter(
    is.finite(cases_per_mil),
    is.finite(deaths_per_mil),
    cases_per_mil > 0,
    deaths_per_mil > 0
  ) %>%
  lm(
    log(deaths_per_mil) ~ log(cases_per_mil),
    data = .
  )

summary(global_mod)
```

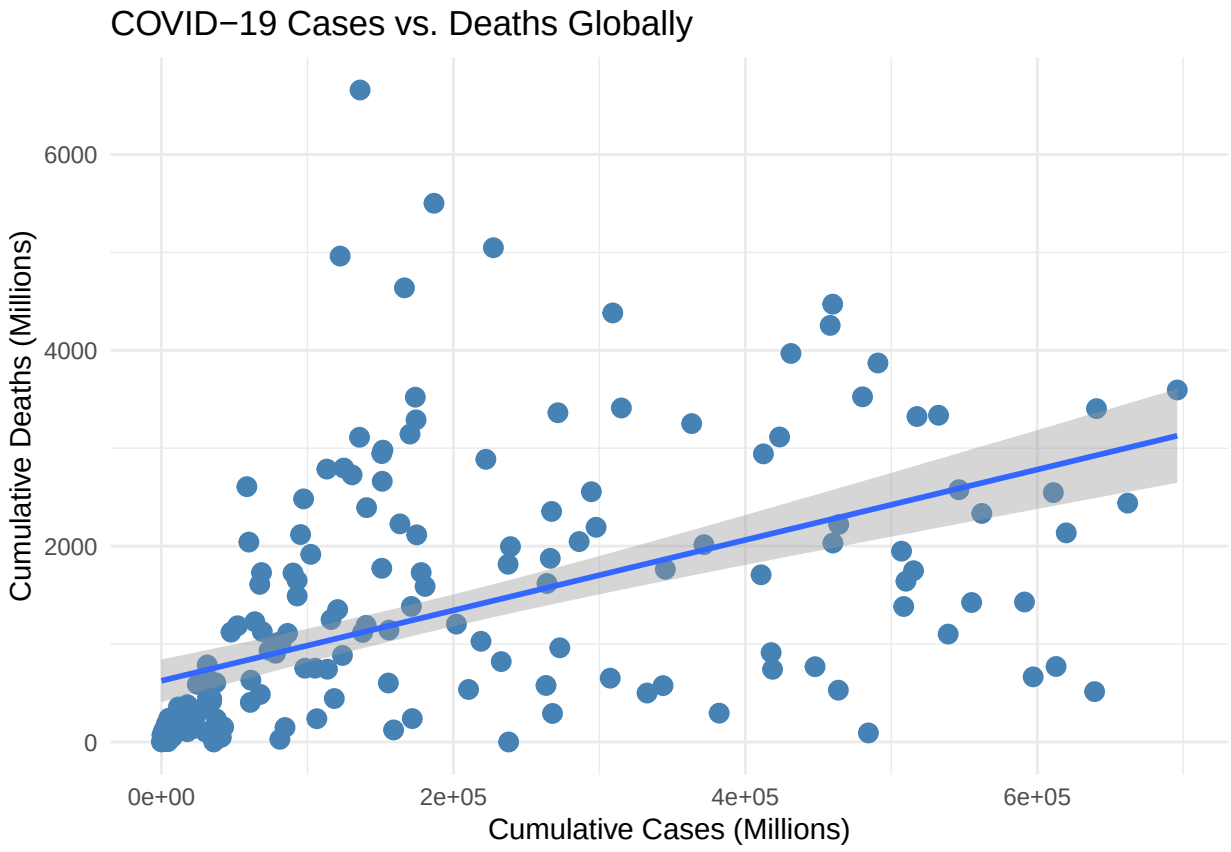
```
##
```

```
## Call:
## lm(formula = log(deaths_per_mil) ~ log(cases_per_mil), data = .)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.3534 -0.5177  0.0886  0.6189  2.0497
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)         -1.2723     0.3255  -3.909 0.000128 ***
## log(cases_per_mil)   0.6878     0.0294  23.396  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8989 on 192 degrees of freedom
## Multiple R-squared:  0.7403, Adjusted R-squared:  0.739
## F-statistic: 547.4 on 1 and 192 DF,  p-value: < 2.2e-16
```

```
global_pred <- global_country_totals %>%
  mutate(
    pred_global = predict(global_mod, newdata = global_country_totals)
  )
```

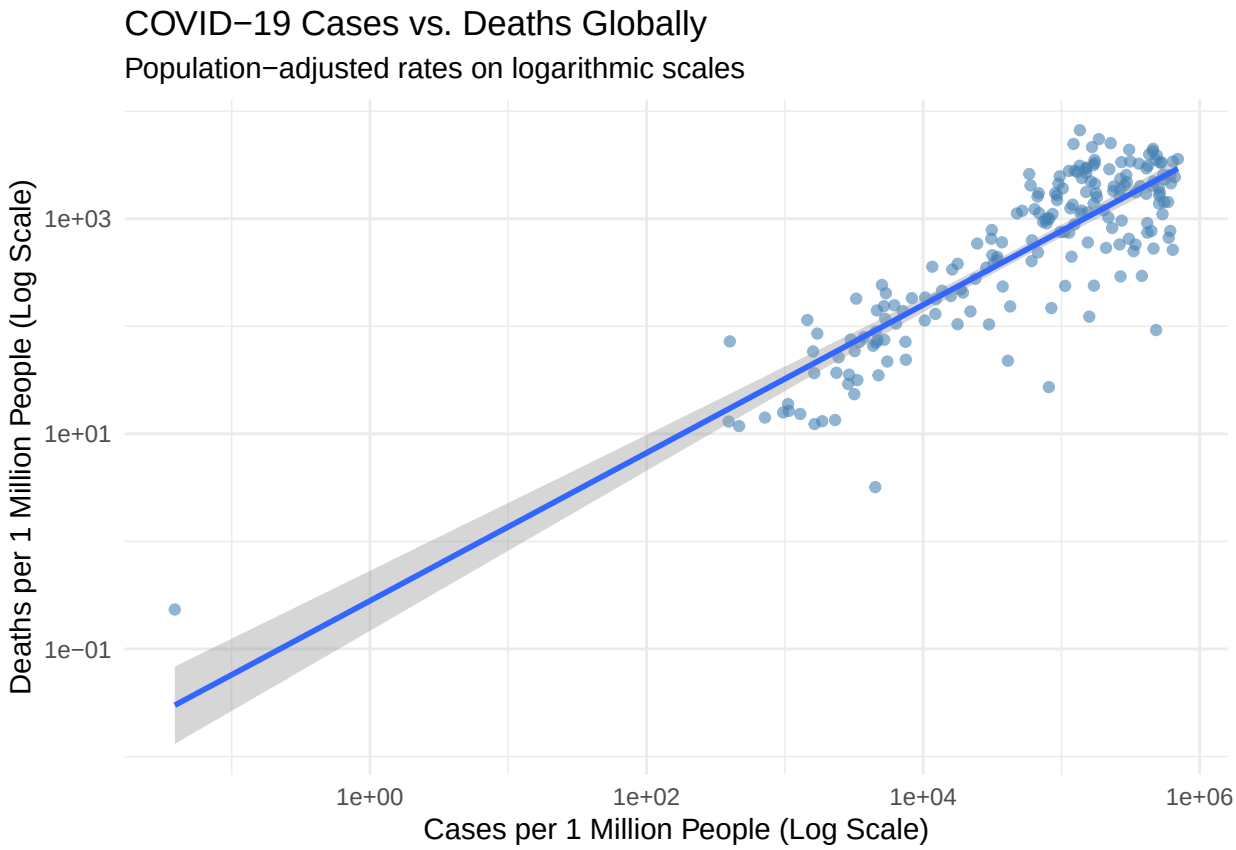
```
global_country_totals %>%
  ggplot(aes(x = cases_per_mil, y = deaths_per_mil)) +
  geom_point(color = "steelblue", size = 3) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
  title = "COVID-19 Cases vs. Deaths Globally",
  x = "Cumulative Cases (Millions)",
  y = "Cumulative Deaths (Millions)"
) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

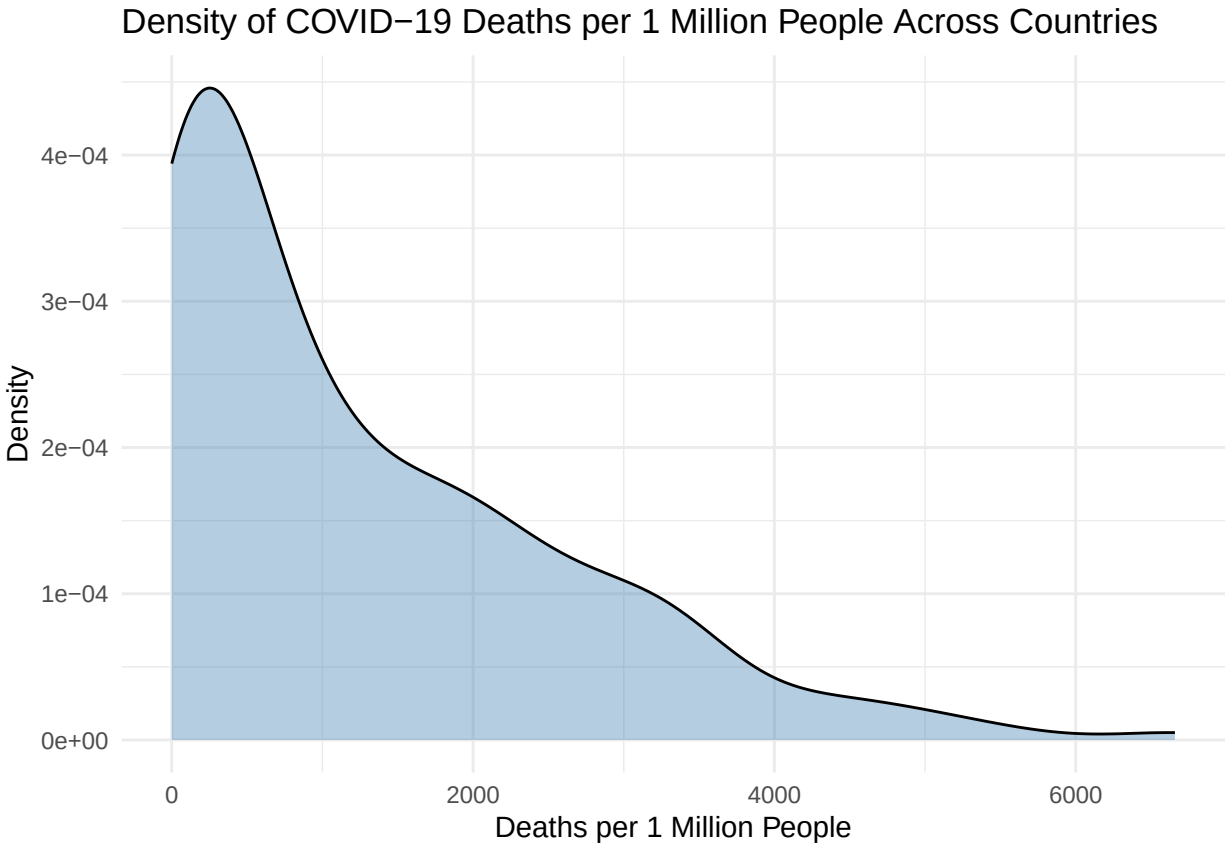



```
global_country_totals %>%
  filter(
    is.finite(cases_per_mil),
    is.finite(deaths_per_mil),
    cases_per_mil > 0,
    deaths_per_mil > 0
  ) %>%
  ggplot(aes(
    x = cases_per_mil,
    y = deaths_per_mil
  )) +
  geom_point(color = "steelblue", alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE) +
  scale_x_log10() +
  scale_y_log10() +
  labs(
    title = "COVID-19 Cases vs. Deaths Globally",
    subtitle = "Population-adjusted rates on logarithmic scales",
    x = "Cases per 1 Million People (Log Scale)",
    y = "Deaths per 1 Million People (Log Scale)"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
global_country_totals %>%
  filter(deaths_per_mil > 0) %>%
  ggplot(aes(x = deaths_per_mil )) +
  geom_density(fill = "steelblue", alpha = 0.4) +
  labs(
    title = "Density of COVID-19 Deaths per 1 Million People Across Countries",
    x = "Deaths per 1 Million People",
    y = "Density"
  ) +
  theme_minimal()
```



Anaylsis: Global Linear Regression The linear regression model uses global new cases (Millions) to predict new deaths (Millions). The relationship between new cases and new deaths on the global scale is much more linear than in the United States alone. The adjusted R-squared of the simple linear regression model is 0.7847, showing that the model explains about 78% of the variability. The scatterplot with deaths and cases per million featured a fan-shape, because of this we used logarithmic analysis, where both variables were transformed using logarithmic scales to reduce skewness and improve the visualization relationship. The plot shows a positive association between reported cases rates and death rates. The density plot shows a skewed curve with the majority of the deaths being below 1500-2000 deaths per 1 Million people.

Discussion of Results

How did reported COVID-19 cases and deaths change over time in the United States and globally between 2020 and March 2023? The cases and deaths had a sharp increase in 2020 both in the United States and globally. While the new cases and new death are mostly parallel, they start to slightly diverge with less new deaths per new cases a year.

The annual deaths-to-cases percentage was calculated by dividing newly reported deaths by newly reported cases for each year. The results indicate that the percentage declined from 2.27% in 2020 to 1.74% in 2021 and 0.33% in 2022. This represents an approximate 85.5% relative decrease between 2020 and 2022. The number of reported cases increased substantially in 2022, but the number of reported deaths decreased compared to 2021. This suggests that over the coarse of the pandemic the relationship between cases and deaths changed. 2023 cannot be compared because the data does not include the entire year.

Many factors may have contributed, including vaccination, infection immunity, and the virus variant in circulation, treatment, and testing/reporting practices. The annual reported deaths to cases are not equivalent

to infection fatality rates since deaths may be recorded weeks after the cases. In addition, many cases go unreported.

How did reported COVID_19 death rates per 1000 people different among U.S. states and countries?

U.S Comparisons Arizona had the highest reported COVID-19 death rate among the U.s.s states & territories examined, with 4.55 deaths per 1,000 people. Oklahoma & Mississippi followed with 4.54 and 4.49 deaths per 1,000 people. Among the states with lowest reported death rates were Hawaii (1.3), Vermont (1.49), and Utah (1.65). Some U.s. island territories reported even fewer deaths per cases. These results show variation across states in reporting COVID-19 deaths. These differences may be because of demographics, underlying health conditions, vaccination rates, healthcare access and local public health responses.

Global Comparisons There are large variations across the globe with reported COVID-19 death rates. Peru had the highest reported death rate with approximately 6,658 deaths per million people (equivalent to 6.6 people per thousand, if comparing to the same scale used in the U.s. data). Bulgaria and Hungary followed with approximately 5.5 and 5.05 deaths per 1,000 people. Countries with the lowest reported death rates were Benin, Burundi, Chad, and North Korea. However, these results should be interpreted cautiously because lower reported mortality does not necessarily indicate actual deaths.

The results indicate substantial variation in population-adjusted COVID-19 reported death rates in the United States and the around the world. These differences cannot be attributed to a single cause. Additional information about demographics, vaccination rates, healthcare capacity, and reporting practices would be needed to gain further insight into explaining the variations.

What is the relationship between reported COVID-19 cases and deaths and how well can regression explain that relationship?

The analysis examined the relationship between cumulative reported COVID-19 cases and deaths across the countries using population-adjusted rates per million people. A log-linear regression model was used to evaluate the association between the two variables. The scatterplot shows a positive relationship between the reported cases and death rates. Countries with higher levels of reported cases also had higher levels or reported deaths, generally. The original plot showed considerable variation and a fan-shaped pattern. After applying logarithmic transformations, the relationship appeared more linear and making the model a more useful approach. The adjusted R-squared of the simple linear regression model is 0.7847, showing that the model explains about 78% of the variability. However, variation still exist among countries of similar case rates. These differences may reflect variations in demographics, vaccination rates, healthcare capacity, and reporting practices. Although regression can identify and summarize statistical relationships, it cannot establish that the differences along caused the observed difference in death rates.

Data Bias and Limitations

Underreporting of COVID-19 Cases

Not all infections were officially reported. Individuals with mild or no symptoms may not have been tested. The increased ability for at-home testing may have reduced the number of cases reported.

Reporting Practice Variation

Countries and states may use different testing strategies and reporting schedules. These differences can affect comparisons between geographic locations.

Population Differences

Comparing total cases and deaths without accounting for population size can be misleading. This analysis uses population-adjusted death rates to make comparisons more meaningful. The rates do not account for differences in demographics such as age, health conditions or healthcare access.

Regression Model Limitations

The regression models examine statistical relationships between the reported cases and the reported deaths. Relationships do not indicate causation. Other variables such as vaccination rates, healthcare access, demographics, treatment, and testing practices may influence death rates.

Conclusion

This project demonstrates the value of using data visualization, population-adjusted comparisons and regression modeling to explore COVID-19 outcomes. The results show that the relationship between reported cases and deaths changed over time and that the mortality rates varied substantially across the world. This analysis showed that reported deaths declined relative to reported cases over the course of the pandemic. The regression analysis also showed a positive relationship between reported cases and reported deaths. Further research that incorporates vaccination rates, healthcare access, demographics, treatment, and testing practices could provide more insight into the drivers of variance between the cases and death rates.

References

"COVID-19 Data Repository by the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University" or "JHU CSSE COVID-19 Data" for short, and the url: https://github.com/CSSEGISandData/COVID-19.

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